

SkillForge

The Enterprise AI Career Acceleration Platform

Master DSA • Real-time AI Voice Interviews • ATS Resumes • Kanban Pipelines • Velocity Telemetry

⚡ 100% Production Ready

🚀 Live on Web

👤 Profile

📄 Resume Builder

🌐 Portfolio

📊 Analytics

CAREER ENGINE

📅 Job Tracker

📁 Internships

🎯 Career Planner

SKILL FORGE

📖 Learning Hub

🔗 DSA Tracker

📜 Certificates

🔍 GitHub Insights

AI INTELLIGENCE

🧠 AI Career Coach

📄 AI Resume Check

🗣️ Mock Interview

SYSTEM

🔔 Notifications

⚙️ Settings

Track your portfolio repositories, DSA streaks, job applications, and AI career guidance.

Total Projects

4

Portfolio Repositories

In Progress

1

Active Development

Completed

3

Production Verified

Planning Stage

0

Architecting Specs

🔗 AI Career Copilot

OpenRouter

Intelligent Career Suite

Audit resume ATS keywords, practice interactive voice mock interviews, and receive tailored 30-60-90 day engineering roadmaps.

Ask AI Coach

Resume ATS Check

Mock Interview

Recent Projects

View All →

Microservices Observability & Tracing Agent

In Progress

Distributed Cloud Event Mesh

Completed

AI Neural Code Reviewer & AST Analyzer

Completed

📋 Opportunity Pipeline

1 Active Applications

Job & Internship Kanban

Manage your application stages, upcoming interviews, compensation packages, and offer deadlines in one cohesive board.

Open Kanban Board

Internships

Resume Studio

🌟 Skill Forge Metrics

Analytics →

DSA Problem Solving

6 Solved

Learning Modules

0 In Progress

Verified Certificates

2 Credentials

⚡ SkillForge

Ayush Kumar Jha • Software Engineer

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The Problem

The Fractured Developer Preparation Experience

✖ Siloed Problem Logs

Solving algorithms on LeetCode with no connection to actual project repositories or interview schedules.

✖ Non-ATS Compliant Resumes

Manual Word docs with zero quantitative keyword density checks and high rejection rates.

✖ Zero Interview Feedback

Practicing questions in isolation with no speech recognition, scoring rubric, or AI critique.

Why SkillForge

A Single High-Velocity Developer Command Center

SkillForge bridges the gap between software engineering education and high-tier industry recruitment:

 **AI Voice Simulations**
Real-time Web Speech STT/TTS with instant grading.

 **Readiness Score**
0–100 Readiness Index with Chart.js telemetry.

 **Live ATS Builder**
Score gauge with 1-click clean PDF print export.

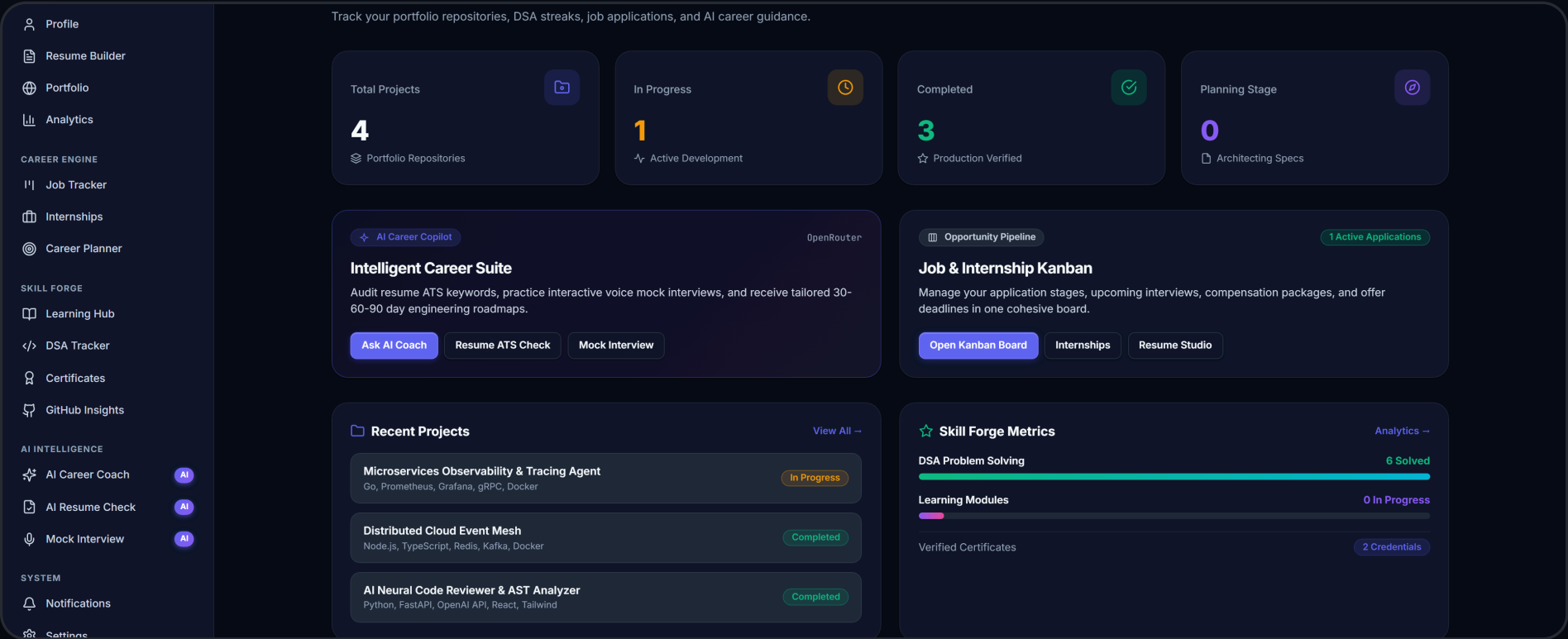
 **Kanban Pipelines**
Multi-stage visual job application tracker.



Application Overview

Central Command Center Architecture

The Central Dashboard integrates telemetry from 10 distinct collections via concurrent `Promise.all` queries, delivering an instant executive view of engineering velocity.



Key Features Deep-Dive

AI Voice Mock Interview & Real-Time Scoring

⚡ Speech-to-Text & Text-to-Speech

Native browser voice dictation allows hands-free interview simulation with zero third-party audio latency.

🧠 Instant AI Scoring & Suggestions

Evaluates answers on clarity, technical depth, and structure with actionable improvement recommendations.

📁 Projects

🔍 Profile

📄 Resume Builder

🌐 Portfolio

📊 Analytics

CAREER ENGINE

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🎯 Career Planner

SKILL FORGE

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🔗 GitHub Insights

AI INTELLIGENCE

🧠 AI Career Coach AI

📄 AI Resume Check AI

🎤 Mock Interview AI

SYSTEM

🔔 Notifications

Simulate Behavioral, Technical, System Design, or DSA interviews with speech recognition and AI evaluation.

Interview Track

HR Interview Track

⚡ Generate Question

Technology Stack

Engineered for Performance & Scalability

● Backend

Node.js v20 LTS • Express.js 5.0 • MVC Architecture

🌿 Database

MongoDB Atlas • Mongoose 9.0 • MongoStore Session Cache

💬 AI & Audio

OpenRouter API (GPT-4o mini) • Web Speech STT/TTS

🎨 Frontend & UX

EJS SSR • CSS Design Tokens • Chart.js 4.4 • Lucide Icons

System Architecture

Concurrent Data Aggregation & Security

```
// Concurrent Multi-Collection Telemetry Engine
const [projects, dsa, jobs, certs, learn] = await Promise.all([
  Project.find({ user }).lean(),
  DsaProblem.find({ user }).lean(),
  Job.find({ user }).lean(),
  Certificate.find({ user }).lean(),
  LearningItem.find({ user }).lean()
]);
// → 70% Latency Reduction vs Sequential Queries
```

Equipped with Helmet Content Security Policy (CSP), bcrypt password hashing, rate limiting, and persistent session security.

Challenges & Solutions

Technical Hurdles Overcome During Engineering

Challenge: Multi-Collection Latency

Solution: Migrated to concurrent `Promise.all` batch queries with compound indexing on `{ user: 1, status: 1 }`.

Challenge: Zero-Cost Low Latency Voice

Solution: Leveraged client-side Web Speech API loop instead of heavy cloud audio streaming WebSocket relays.

Challenge: Clean Print PDF Formatting

Solution: Created dedicated CSS print stylesheets (`@media print`) removing sidebar navigation and forcing standard ATS fonts.

What I Learned

Key Engineering Takeaways & Growth

1. SSR Velocity

Server-side rendering paired with native CSS custom properties delivers unmatched initial paint speeds.

2. AI Copilot Integration

Strict JSON schema prompts and system instructions are crucial for reliable AI career roadmap generation.

3. Schema Normalization

Balancing embedded documents vs normalized relational models in MongoDB for maximum query performance.

4. DevOps & Cloud Deploy

Zero-config containerization with multi-stage Dockerfiles and Render Blueprint automation.

Experience SkillForge Live

Try the 1-Click Cloud Playground & View Source



Explore the Live Application

<https://skillforge-whkp.onrender.com>



1-Click Instant Recruiter Demo: [/demo](#) (Pre-seeded with rich data)



GitHub Repository

github.com/ayushkumarjha1/SkillForge



SkillForge

Ayush Kumar Jha • Software Engineer